

Topic: Multilingual Real-Time Sign Language Translator

Mixture-of-Experts (MoE) Deep Learning System for Sign Recognition & Translation

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Previous Review Feedback & Project Solution

Title: Multilingual Real-Time Sign Language Translator

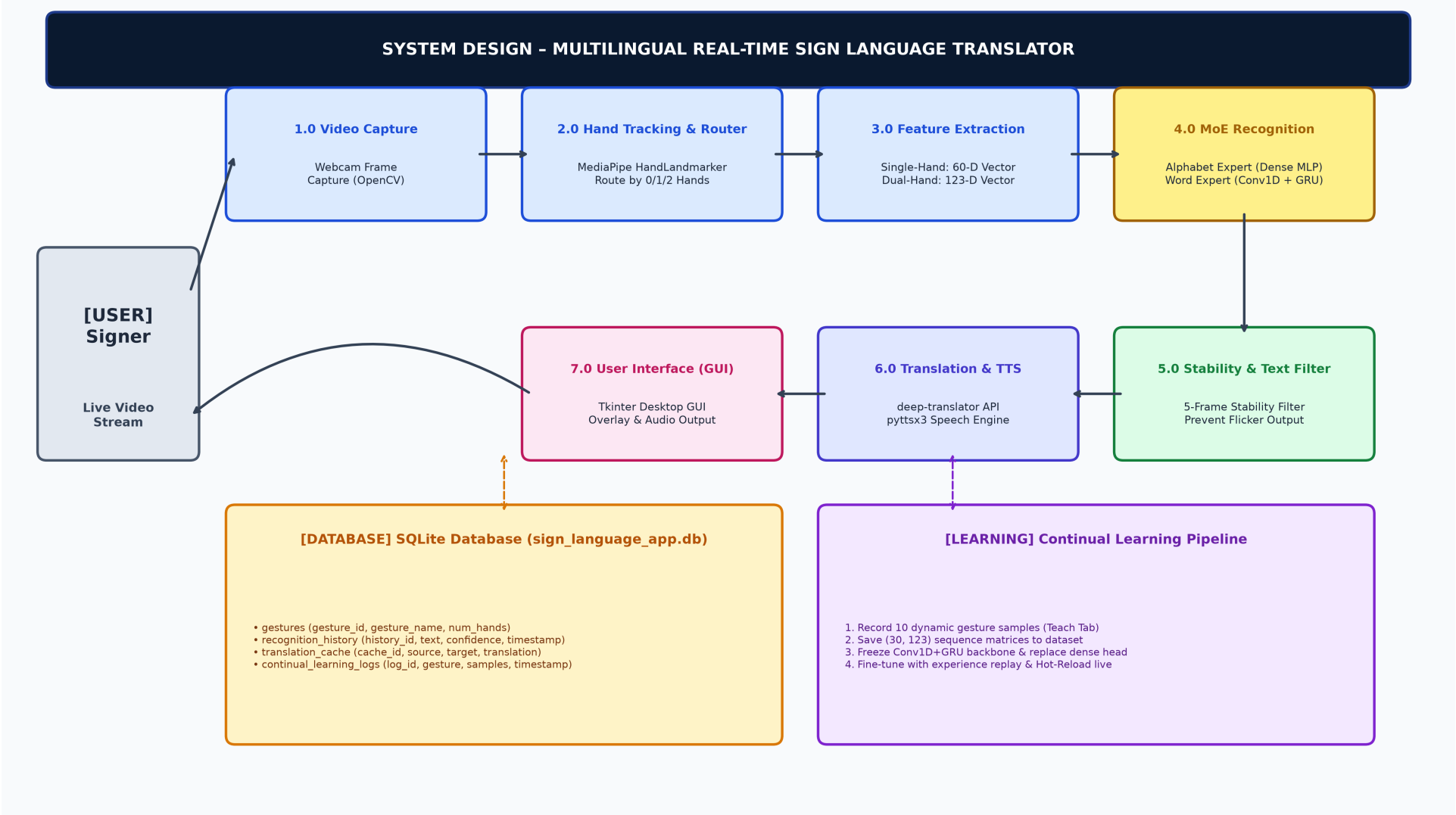
Problem Definition: Communication between sign language users and non-signers is difficult due to language barriers and limited real-time translation tools.

Proposed Solution: Develop a real-time system utilizing a Hierarchical Mixture-of-Experts (MoE) deep learning architecture that recognizes static letters (A-Z) and dynamic word gestures, translating them into text and speech in multiple languages.

Objectives:

- Recognize sign language gestures accurately in real time with an MoE router (0/1/2 hands).
- Translate recognized signs into multiple target languages (Kannada, Hindi, Tamil, Telugu, etc.) with 0ms cache lookup.
- Provide synchronized text overlay and audio speech (TTS) output.
- Support live continual learning to extend the dictionary with custom gestures without full retraining.

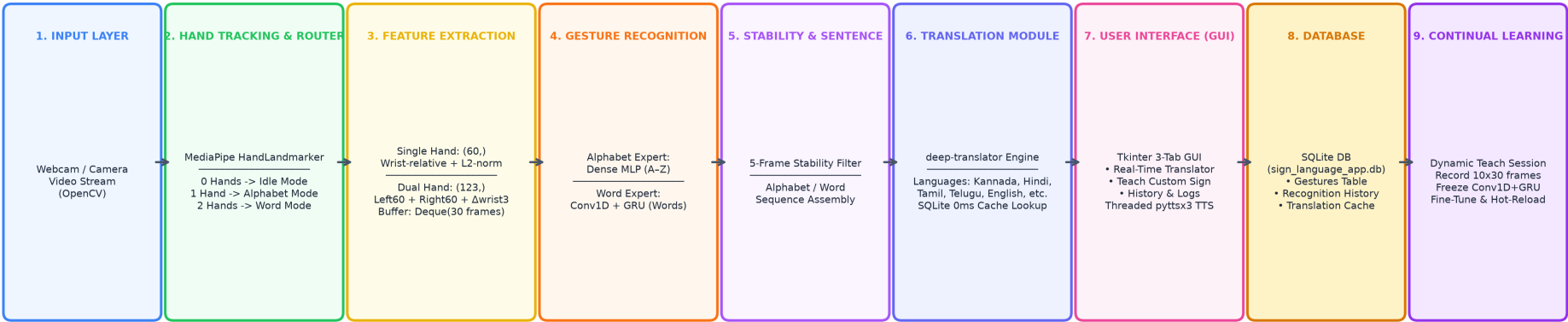
System Design Diagram



System Architecture Diagram

SYSTEM ARCHITECTURE DIAGRAM

Multilingual Real-Time Sign Language Recognition and Translation System (MoE Edition)



SUMMARY OF SYSTEM ARCHITECTURE & CODEBASE MODULES

- Webcam / OpenCV Captures Frames -> MediaPipe Tasks extracts 21 3D landmarks per hand.
- MoE Router dispatches 1-hand input to Alphabet Expert (Dense MLP, 60-D features) and 2-hand input to Word Expert (Conv1D + GRU, 123-D x 30 frames).
- 5-Frame Stability Filter ensures smooth, flicker-free gesture recognition before trigger.
- Translation Module uses deep-translator with SQLite Cache (0ms latency for repeated phrases) & pyttsx3 for offline audio TTS.
- Tkinter Desktop Application handles real-time visual overlay, translation selection, history viewing, and dynamic continual learning.

Methodology

Step 1: Data Collection & Input — Collect static images (ASL A-Z) and dynamic video sequences (WLASL word dataset).

Step 2: Hand Landmark Tracking — Detect 21 3D hand landmarks per hand using MediaPipe HandLandmarker.

Step 3: Feature Engineering — Extract 60-D features for single hand, and 123-D features across 30 frames for dual hands.

Step 4: MoE Routing & Recognition — Route single hand to Alphabet Expert (MLP) and dual hands to Word Expert (Conv1D+GRU).

Step 5: Stability Filtering & Sentence Assembly — Apply 5-frame stability filter to eliminate flicker.

Step 6: Multilingual Translation & TTS — Translate text via deep-translator with SQLite cache (0ms lookup) & pyttsx3 speech.

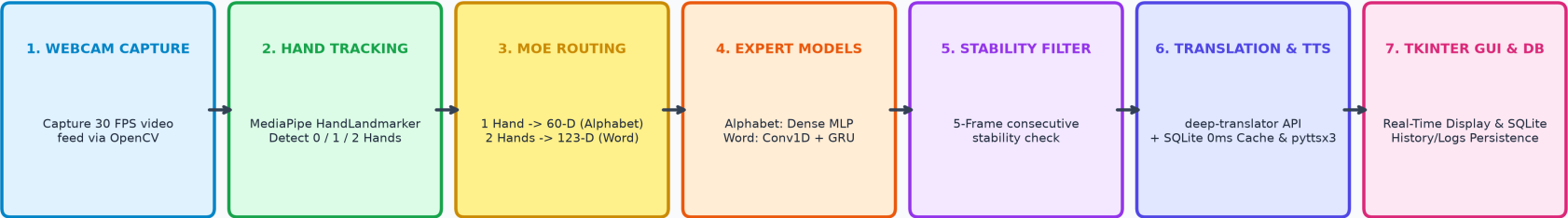
Step 7: Continual / Transfer Learning — Record 10 custom gesture samples in GUI, freeze Conv1D+GRU, fine-tune dense head & hot-reload.

Step 8: Output & Persistence — Display real-time UI overlay, play speech, and save history to SQLite DB.

Methodology & Implementation Pipeline Diagram

Methodology & Implementation Pipeline

Multilingual Real-Time Sign Language Translator — Mixture-of-Experts (MoE) Architecture



Thank You!